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Measuring Polycentricity and its Promises

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RESEARCH BRIEFING

Measuring Polycentricity and its Promises

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ABSTRACT *Even a decade after the concept of polycentric development became popular and increasingly widespread in Europe as a normative policy stance allegedly leading to cohesion and competitiveness, its empirical basis is still rather weak. This is partly due to a lack of conceptual clearness, which makes its measurement difficult. This research briefing synthesises the results of two recent ESPON projects that aim to create a quantitative measurement of the extent of polycentricity of national urban systems, as well as the links they find between polycentricity and economic and social objectives. Both approaches have their limits and have been, and can be criticized from various perspectives. Some suggestions on how to proceed with this research agenda are presented.*

Introduction

The concept of polycentric development has gained widespread currency in planning and territorial development strategies, though it remains a rather fuzzy concept as it seems to mean different things to different actors and on different scales (Kloosterman & Musterd, 2001; Davoudi, 2003; Hague & Kirk, 2003). The scale on which the concept is applied ranges from individual cities (e.g. Roberts *et al.*, 1999) to city regions (e.g. Hall & Pain, 2006; Parr, 2004; Meijers, 2007a), to nation-states (Waterhout *et al.*, 2005) and, finally, to the European scale (see CEC, 1999). Davoudi (2003) observes that as the scale increases, the concept of polycentricity becomes gradually less analytical, and instead becomes the expression of a normative agenda. Indeed, polycentricity is not just a descriptive term, but also a policy stance prescribing a means to promote and equalize economic growth (Hague & Kirk, 2003). Its main promise is that it would link the seemingly conflicting goals of greater cohesion and competitiveness (Waterhout *et al.*, 2005). The theoretical basis for this, however, is weak, as for instance regional growth

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theories such as the New Economic Geography hardly take the shape of the urban system into account (Meijers & Sandberg, 2008).

The European Spatial Development Perspective (ESDP) presents as a major objective the development of a balanced and polycentric urban system, as the economic fabric of Europe is dominated by one large metropolitan system mainly located in the northwest corner of the continent: the pentagon (CEC, 1999; Krätke, 2001; Copus, 2001). It is argued that “the economic potential of all regions of the EU can only be utilized through the further development of a more polycentric European settlement structure” (CEC, 1999, p. 20). The objective of polycentric development is currently being taken further in the current debate about a European territorial cohesion policy (Faludi, 2005, 2006). The EU member states recently agreed on a “Territorial Agenda of the European Union” (EU Ministers responsible for Spatial Development, 2007), which again underlines the importance of strengthening polycentric development. The substance of the concept has not changed much since the ESDP, as it is still about the development of multiple core areas next to one existing core. However, it appears that polycentric development is now relatively more concerned with cohesion rather than competitiveness objectives.

Alongside the European scale, many European countries also individually pursue a polycentric development policy, often addressing the dominance of their prime city to diminish regional disparities (see Waterhout *et al.*, 2005 for an overview). A few examples include France, which has a long tradition of combating the dominance of Paris (see Guigou, 2000; Baudelle & Peyrony, 2005), while Ireland tries to develop “gateway cities” in lagging regions in order to overcome its overdependence on Dublin (DELG, 2002; Davoudi & Wishardt, 2005). This does not mean that in all countries where the polycentricity concept has been used it has become influential in the sense of directing the application of concrete policy instruments. What can be seen, however, are remarkable examples of application in spatial planning documents—spatial visions and perspectives—although at the same time implementation is often a weak point.

Within the ESPON (European Spatial Planning Observation Network) framework, there have also been attempts recently to measure polycentricity on the scale of countries and the EU, thus implementing a more analytical than normative approach to polycentric development on these scales (ESPON 1.1.1, 2002–2004; ESPON 1.4.3, 2006). This is of great importance as it allows for an empirical assessment of the many claims that have been made for the concept of polycentricity on these scales. The empirical assessment of the polycentricity concept has become all the more urgent now that polycentricity has won itself this prominent place in many national and European spatial development strategies, which often involve substantial budgets (most notably the European structural and cohesion funds). The possible effectiveness of such policies is also determined by the accuracy of the assumptions on which they build. Therefore, it is both surprising and alarming that a key concept such as polycentric development and its promises of cohesion and competitiveness has so far hardly been assessed empirically on these higher spatial scales.

Part of the explanation for this empirical deficit is the lack of a common and shared understanding of what polycentricity is and how it should be properly measured. One of the major reasons why the concept of polycentricity is still rather vague is that it is sometimes associated with both morphological aspects and functional relationships between cities. As will be explained further on, it is highly advisable to reserve the term for morphological issues and to distinguish it from functional networks between cities. So

far, the lack of a shared understanding stands in the way of an assessment of its relation with aspects of cohesion and competitiveness. However, two research projects carried out within the ESPON framework have tried to shed light on these matters. The first is the ESPON 1.1.1 project on “Urban areas as nodes in a polycentric development”. The second is the ESPON 1.4.3 project, “Study on Urban Functions” that aims to improve the ESPON 1.1.1 findings. As is common for this early stage of exploration, it appears that researchers have so far not reached consensus on the right approach to measure polycentricity.

In this paper, we first observe the outcomes of both projects. This refers both to their analysis of the extent of polycentricity of European countries, as well as the relationships they have explored in terms of economic competitiveness, cohesion and sustainability. These outcomes are strikingly different, and the section ‘Explaining Differences’ provides explanations for this. Finally, based on the pros and cons of both approaches to measuring polycentricity, some suggestions will be presented on how to proceed with this important research agenda.

Measuring Polycentricity and its Promises

The ESPON research programme aspires to achieve a better understanding of spatial trends, problems and opportunities on a European scale, to monitor them using newly developed indicators and mapping techniques and to provide tools and instruments for a better territorial governance at EU level. The ESPON 1.1.1 project was among the first projects that started in 2002, and focused on “the role, specific situation and potentials of urban areas as nodes in a polycentric development”. Within this comprehensive project an analysis of the extent of polycentricity and monocentricity of countries was made. The project was led by Nordregio, while by the end of the project (2004), 14 research organizations from across Europe were involved as main or consultative partners. Carried out at a later stage (2006), the ESPON 1.4.3 project aimed to take forward the analysis of project 1.1.1, in particular the delimitation of functional urban areas (FUAs) and the analysis of polycentricity building on that delimitation. The project was led by IGEAT/Free University of Brussels and the team consisted of two other partners and one other expert. The final report was published in March 2007.

Table 1 presents the results of ESPON 1.1.1’s and ESPON 1.4.3’s analysis of the extent of mono- or polycentricity (both are considered extremes on the same scale) of European countries. Higher scores indicate more polycentricity. There are remarkable differences between their outcomes. In fact, they do not even statistically correlate. For instance, according to ESPON 1.4.3, countries such as Germany, Spain, Italy, the Netherlands, Lithuania, Poland, Romania, the Slovak Republic and most of the Nordic countries are much more polycentric than according to ESPON 1.1.1, while Cyprus, Denmark, Greece, Ireland, Latvia and Slovenia turn out to be much more monocentric.

These scores on polycentricity allow for an assessment of the hypothesized effects of polycentricity. Although not a prime objective, ESPON 1.1.1 briefly explored the links between the extent of polycentricity of European countries and three major objectives of the EU: economic competitiveness, social equity and environmental sustainability. Proxies for these were GDP/capita, the gini coefficient of the GDP per capita measured at the nuts 3 level and total energy consumption for transport

Table 1. The extent of polycentricity of European countries according to ESPON 1.1.1 and ESPON 1.4.3

Project	ESPON 1.4.3	ESPON 1.1.1	ESPON 1.1.1 sub indexes		
	Level of Polycentricity	Polycentricity index	Size index	Location index	Connectivity index
Austria	56.8	57.4	63.3	39.3	77.1
Belgium	69.6	70.3	86.6	60.5	67.1
Bulgaria	73.2	68.5	77.1	80.2	52.6
Switzerland	75.4	66.6	82.9	57.9	62.3
Cyprus	51.4	87.3	75.7	100.0	89.1
Czech Republic	74.2	63.6	79.2	51.7	63.5
Germany	93.4	71.2	86.4	56.1	75.2
Denmark	56.6	72.5	71.6	90.9	59.3
Estonia	45.6	54.3	64.7	94.8	26.4
Spain	77.6	53.6	81.6	30.7	62.3
Finland	63.6	49.1	73.9	32.1	50.6
France	62.4	67.6	66.4	77.3	60.9
Greece	38.6	63.4	36.6	95.9	73.6
Hungary	54.8	56.1	61.6	57.7	50.4
Ireland	45.8	76.1	63.1	100	70.6
Italy	86.2	66.3	87.5	52	65
Lithuania	72	48.9	76.5	83.5	18.5
Latvia	32.2	56.3	35.5	97	52.4
Netherlands	90.8	72.2	86	60.2	73.8
Norway	66.4	44.4	75.1	22.3	52.7
Poland	85.6	74	84.1	83.1	58.7
Portugal	54.4	58.3	49	55.8	73.3
Romania	77.2	66.3	78.3	80.9	46.6
Sweden	71.8	58.9	80.4	37.3	69
Slovenia	56.8	79.1	76	91.6	72
Slovakia	74.8	64.2	83.5	77	41.6
United Kingdom	63.8	66.8	77.3	55.5	70.6

Sources: ESPON 1.1.1, 2002–2004; ESPON 1.4.3, 2007.

respectively. Some correlations were found. More polycentric countries tend to be associated with a higher GDP/capita and with the use of less energy for transport per unit. However, these countries also tend to have higher income differences within countries. These correlations are stronger for the old EU-15 than for the recently accessed countries.

As with the scores on polycentricity, ESPON 1.4.3 presents quite opposite results. They explore the relationships between the level of polycentricity and the development of the GDP/capita relative to the European average for different time spans. No significant correlation was found, but from the direction of the correlation it appears that the more mono-centric countries score slightly better. According to the ESPON 1.4.3 team, this is in line with trends towards globalization that favour the main advanced services nodes operating on a global scale. Also, it conforms to the findings of Meijers and Sandberg (2008), who found no evidence of a polycentric urban system being correlated with less regional disparities within countries.

Explaining Differences

These differences raise questions about the value that should be attached to the outcomes. How is it possible that two research projects that aim to measure a similar feature of urban systems and its effects arrive at such different conclusions? Here, we provide two main explanations. The first concerns the way polycentricity is defined. The second concerns the different emphasis that is placed on the elements that define polycentricity.

The Definition of the Concept of Polycentricity

ESPON 1.1.1 departs from the observation that polycentricity is an ambiguous, multi-scalar and ill-defined normative concept. ESPON 1.1.1 stresses that two structural elements are of particular relevance to polycentricity: morphological elements laying out the size and territorial distribution of urban areas in a given territory, and relational elements, based on the networks of flows and co-operation between urban areas on different scales. Both elements are strongly linked: relations between cities are crucial for polycentricity, as nodes without relations would not form a polycentric system (ESPON 1.1.1, 2004, p. 45). A national urban system is considered more polycentric when the FUAs are more equally sized (see column “size index” in Table 1 for scores), distributed more evenly across the nation’s territory (see column “location index”) and when the accessibility of the FUAs is more equal (see column “connectivity index”). The ESPON 1.4.3 team does not depart from an explicit definition of polycentricity, other than emphasizing that according to them, polycentricity is a purely morphological issue. This implies that both projects depart from a substantially different perspective. The ESPON 1.1.1 project includes the aspect of relationships or flows between the FUAs of the urban system, next to morphological indicators such as the size and spacing of the FUAs. The ESPON 1.4.3 focuses solely on the size distribution of FUAs. Not surprisingly, there is a strong correlation between the size index of ESPON 1.1.1 and the outcomes of the ESPON 1.4.3 project.

Measurement Issues

The strong discrepancies in the outcomes of the analysis in ESPON 1.1.1 and ESPON 1.4.3 also relate to differences in the measurement of polycentricity. Table 2 lists the concrete indicators on which both projects based their measurement of polycentricity. In brackets is the weight given to each indicator in calculating the overall polycentricity score.

Even though both ESPON 1.1.1 and 1.4.3 agree on the fact that the size distribution of FUAs is an important factor in measuring polycentricity, they use different indicators. It is unclear why ESPON 1.4.3 does not build on the ESPON 1.1.1 methodology in this respect. ESPON 1.1.1 starts from the rank-size distribution of FUAs. The more flat this distribution is, and thus the regression line that best fits this distribution, the more polycentric. It would seem that this is a more informative measure than calculating the standard deviation of the population of the FUAs. ESPON 1.1.1’s measure of primacy is also based on the rank-size distribution. There is a difference in the number of FUAs that are included in both analyses. In ESPON 1.1.1 all FUAs with over 50,000 inhabitants, or even just 20,000 inhabitants for some countries, have been included when calculating the rank-size distribution and primacy for each country. ESPON 1.4.3 presents some more differentiation in

Table 2. Indicators used to measure polycentricity in ESPON 1.1.1 and ESPON 1.4.3

ESPON 1.1.1 ^a	ESPON 1.4.3
A. Distribution of sizes FUAs: (33%) 1. The slope of the regression line of the rank-size distribution of FUAs (20%) 2. The degree by which the size of the largest FUA according to population or GDP deviates from the regression line through the rank-size distribution (primacy). ^b (80%)	A. Distribution of sizes FUAs: – Part of the main FUA in the total population of a country (14.3%) – Part of the main FUA in the population of the whole set of FUAs with more than 200,000 inhabitants (14.3%) – Part of the main FUA in the population of the whole set of FUAs with more than 50,000 inhabitants (14.3%) – Average of the differences in population between a FUA and the next smaller FUA in a country, considering the range of FUAs until a lower population threshold of 200,000 inhabitants (14.3%) – Average of the differences in population between a FUA and the next smaller FUA in a country, considering the range of FUAs until a lower population threshold of 50,000 inhabitants (14.3%) – Standard deviation of the population of the set of FUAs with more than 200,000 inhabitants (14.3%) – Standard deviation of the population of the set of FUAs with more than 50,000 inhabitants (14.3%)
B. Spacing FUAs: (33%) 1. The equality in sizes of service areas (based on Thiessen polygons) using the Gini coefficient.	
C. Connectivity: (33%) 1. The slope of the regression line between the accessibility and the population of FUAs (50%). 2. The Gini coefficient of the accessibility of FUAs in a country (50%).	

^a These indicators were developed by Spiekermann and Wegener.

^b These indicators have both been calculated for FUAs that were weighted in terms of population and in terms of GDP (each count for 50%).

thresholds. For unclear reasons, ESPON 1.1.1 gives more weight to primacy than to the steepness of the rank-size distribution.

Towards an Improved Measurement of Polycentricity and its Promises

Both ESPON 1.1.1 and ESPON 1.4.3 can be credited and criticized for several reasons. ESPON 1.1.1 has been ground-breaking in terms of exploring for the first time the issue of polycentricity in quantitative terms. Moreover, it departs from a clear, theoretically founded definition of polycentricity. However, it has been criticized for its outcomes. It appears that most commentators feel that the outcomes for several countries do not

correspond to their tentative impression of the extent of mono- or polycentricity in these cases (see Biot, 2005). Ireland for instance is a case in point. This country came out as one of the most polycentric in ESPON 1.1.1, while many consider it the example *par excellence* of a monocentric country. These concerns were partly the reason to start the ESPON 1.4.3 project. Its outcomes appear more tentatively correct. On the other hand, the project lacks a clear definition of polycentricity and is not very specific about the rationales for using their indicators and their weighting. They also ignore the aspect of the territorial distribution of cities across a nation's territory. The main questions in measuring polycentricity are whether or not this is just a morphological issue and, if so, whether it is only concerned with size or also with the territorial distribution.

Below, some suggestions are presented which aim to arrive at an improved measurement of polycentricity, and subsequently, the more accurate assessment of its promises of cohesion and competition.

A Polycentric Urban System is not a Networked Urban System

Basically, polycentricity refers to the plurality of centres in a given area and is thus by nature a morphological issue. In a synthesis of the defining conditions of a polycentric urban region, Parr (2004) points to, among others, the separation of cities and the size distribution of cities. It appears that morphological characteristics such as the size and spacing of cities are determining factors in establishing whether or not any given area is polycentric or the opposite, monocentric. Following the ESPD terminology and those of many countries' policies, a country can be considered more polycentric when:

- there is less hierarchy in the size of its cities (FUAs), in other words, when the cities are more equally sized and no city dominates the others;
- the cities are more evenly distributed across a country's territory.

In ESPON 1.1.1, the non-morphological dimension of connectivity in terms of potential accessibility was also included. However, a polycentric urban system cannot be equated with a networked urban system. Elsewhere, I have argued that this is also the case on the regional scale, where a polycentric urban region is not necessarily an urban network (Meijers, 2007b). The latter would be the case when features of what has been labelled the network model of spatial organization (see Batten, 1995; Camagni & Salone, 1993; Meijers, 2007c) are present. This includes for instance the features of complementarity between FUAs, a relative disconnection between size and function of FUAs and also the criss-cross orientation of flows. In fact, the extent to which a polycentric system functions as a networked system is a crucial explanation of the performance of this polycentric system. Thus, polycentricity is about morphological features, more particularly the distribution of sizes of FUAs and their territorial distribution. This perspective implies that the ESPON 1.1.1 approach was too broad and the ESPON 1.4.3 approach too narrow.

Focus on the Main Cities

An important explanation for the reason that the ESPON 1.1.1 results, even when ignoring the connectivity aspect, do not seem to correspond to general, tentative impressions of the extent of polycentricity of countries, is that the results are based on too many FUAs. In particular in the larger countries, the calculations for the "flatness" of the urban hierarchy

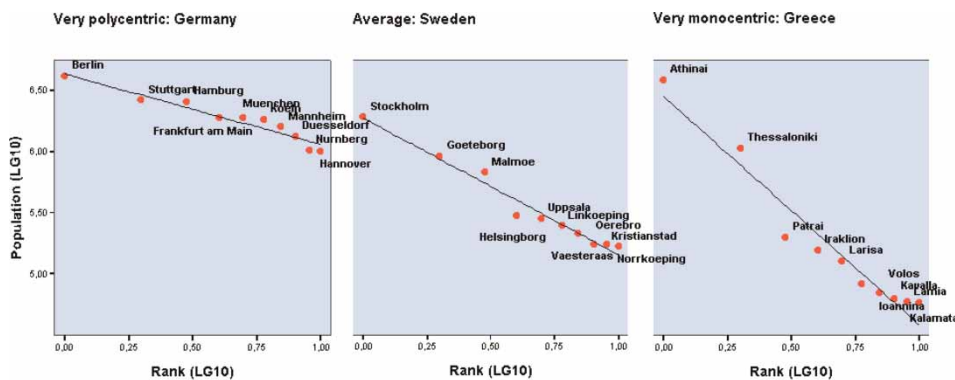


Figure 1. Rank-size distributions of Germany, Sweden and Greece

Source: Meijers & Sandberg, 2008

Slope regression line: Germany -0,571, Sweden -1,118, Greece -1,894

and primacy are strongly influenced by the many smaller FUAs. For instance, does the size of all 253 FUAs in Italy matter for determining its extent of polycentricity? The extent of mono- or polycentricity is generally judged on the basis of the size and spatial distribution of just the handful of largest cities. In general, sample size can be a fixed number of cities, a fixed size threshold, or a size above which the sample accounts for some given proportion of a country's population (see also Cheshire, 1999). The latter has disadvantages, as it turns out that the number of FUAs included in the analysis is large for polycentric countries and small for monocentric countries. Hence, the number of FUAs including some given proportion of the population is in itself an indicator of mono- or polycentricity and applying such a measure twice would distort the picture. A fixed size threshold, as is used in ESPON 1.4.3, is equally less appropriate as in large and more densely populated countries a FUA of say 50,000 inhabitants may be insignificant, while it could be of great importance in small or less populated countries. It could be argued that a city ranked for instance 10th in a country is of importance in that country, despite its possibly small size. Therefore, when measuring polycentricity on the basis of the rank-size distribution, the sample size could best be based on a fixed number of FUAs. The question then is what this number should be. This is open for discussion, but should in any case be limited. However, calculations show that results for a sample size of $n=10$ correlate strongly with sample sizes of $n=5$ or $n=20$ (Meijers & Sandberg, 2008), but differ substantially from the ESPON 1.1.1 findings. Figure 1 gives some examples of calculations of the slope of the regression line using $n=10$. The measurement of primacy should also be calculated relative to a small, but fixed number of FUAs.

Improve Existing Indicators

Some smaller suggested refinements that should be considered when measuring polycentricity include:

- Contrary to ESPON 1.1.1, the largest FUA should also be included in the analysis to calculate the regression line and its slope. However, this largest FUA should be

excluded when the regression line is calculated to measure the primacy (as is done in ESPON 1.1.1).

- The advantage of the ESPON 1.1.1 method of measuring primacy is that, through using the loglinear rank-size distribution, more weight is given to the differences in size between the largest FUAs compared to the weight given to the difference between, say, the 8th and 9th ranked city. ESPON 1.4.3 gives these a similar weight.
- There does not seem to be any reason for not giving the indicator “slope of the regression line” and the primacy the same weight (ESPON 1.1.1 employs a 20/80% weighting).
- ESPON 1.4.3 gives the highest score found a value of 100, and the lowest a value of 0. In such a way, it is possible that a small difference in practice is inflated to a huge one in the calculation. Using standardized z-scores could prevent this.
- As is argued in Nordregio *et al.* (2004), rather than using airline distance to base Thiessen polygons on, these could in a next step be based on travel time and/or costs.
- The size and spatial distribution of FUAs can be substantially different (see Figure 2). For instance, the Netherlands is quite polycentric when looking at the size distribution of cities (using $n=10$), but the main cities are all located in the western region, thus making it rather monocentric from the perspective of the spatial distribution. Some

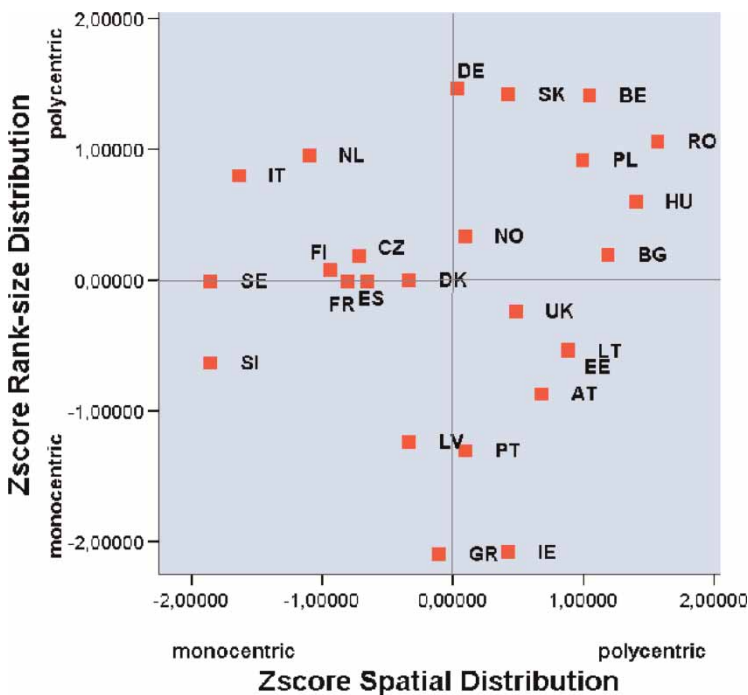


Figure 2. The relationship between polycentricity-indicators based on the rank-size distribution and the spatial distribution of cities across the national territory

Source: Meijers & Sandberg, 2008

To calculate the spatial distribution of FUAs across the territory, Meijers and Sandberg consider the extent to which each NUTS2, or, in smaller countries, NUTS3 region possesses over a major city. Therefore, they consider the spread of a number of FUAs that is similar to the number of NUTS2 (or NUTS3) regions in a country

countries are polycentric by both standards, for instance Belgium, Romania and Poland. It seems informative to explore connections with cohesion and competitiveness also for these subindicators separately rather than solely using their aggregated index.

Final Remarks

Polycentric development has rapidly become the buzzword amongst spatial planners in Europe, both in European policies and at the level of European countries. As is generally the case with new concepts, after the initial policy enthusiasm, the concept of polycentric development needs to be tested on its usefulness and validity if it is to be sustained. This requires a clear definition of polycentricity, subsequently allowing for a proper measurement of the extent of polycentricity. By reviewing and synthesizing approaches to these issues developed in two ESPON projects, this research commentary aimed to further the discussion on this important matter. It appears that there are at present no conclusive answers regarding the question of whether polycentricity is instrumental to achieving economic competitiveness, territorial cohesion and environmental sustainability. Such empirical assessments would provide a welcome contribution to this debate.

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